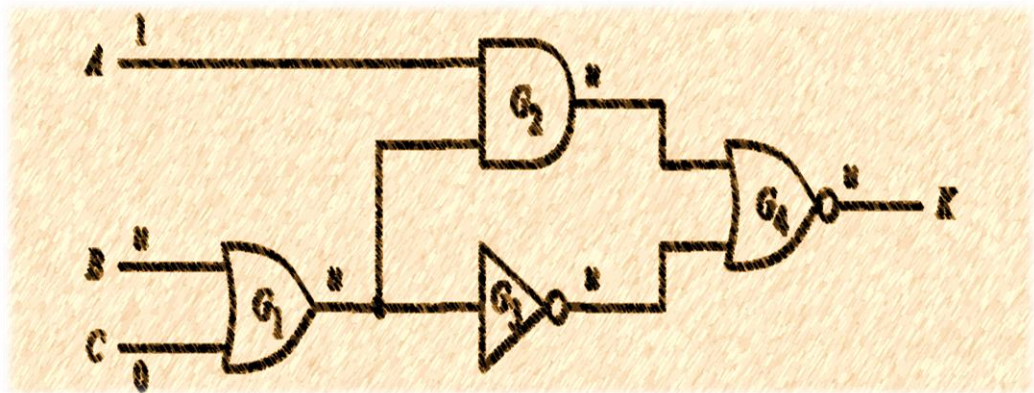


Logic Simulation

- Introduction
- Simulation Models
 - ◆ Logic States
 - ◆ Logic Gate Evaluation
 - ◆ Delay Models
- Logic Simulation Techniques
- Issues of Logic Simulations
- Conclusions



Logic States

- **Binary logic** (0,1)
- **Three-valued logic** (0,1, **u**) aka. **Ternary logic** [Eichelberger 65]
 - ♦ **u** (unknown) for logic value cannot be determined
 - * aka. “x”
- **Four-valued logic** (0,1, **u,z**)
 - ♦ **z** = floating node without conducting path to VDD or GND
 - * aka. **high impedance**
 - ♦ Note: do not confuse “**u**” and “**z**”

AND	0	1	u	z
0	0	0	0	0
1	0	1	u	u
u	0	u	u	u
z	0	u	u	u

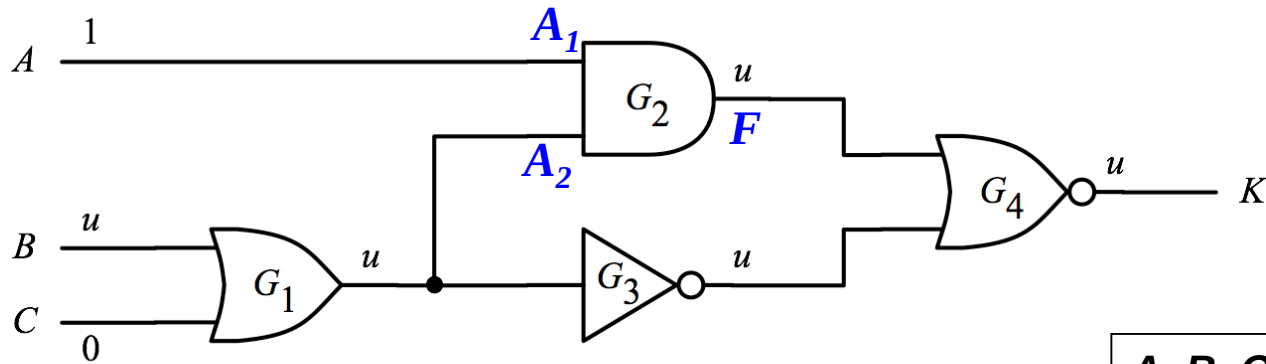
OR	0	1	u	z
0	0	1	u	u
1	1	1	1	1
u	u	1	u	u
z	u	1	u	u

NOT	0	1	u	z
	1	0	u	u

More States, More Accuracy, More CPU Time

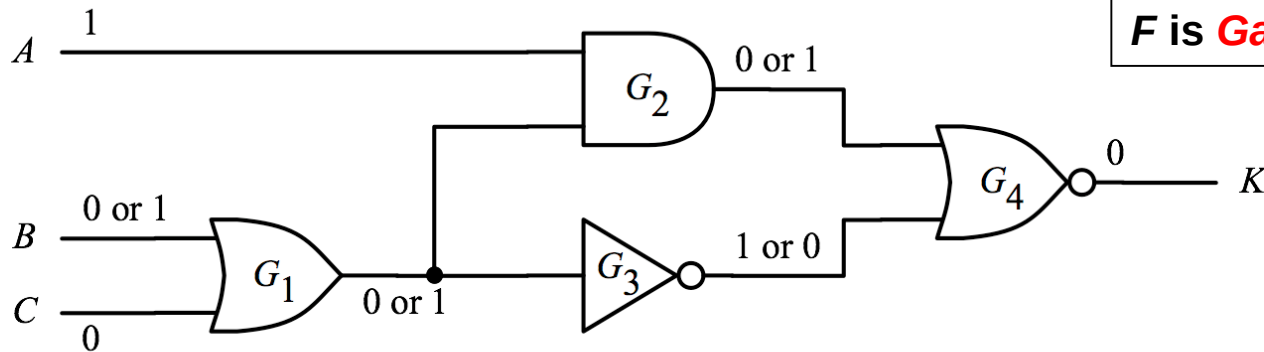
Ternary Logic Simulation

- Problem: output K should be zero



(a) Ternary logic simulation: $K = u$

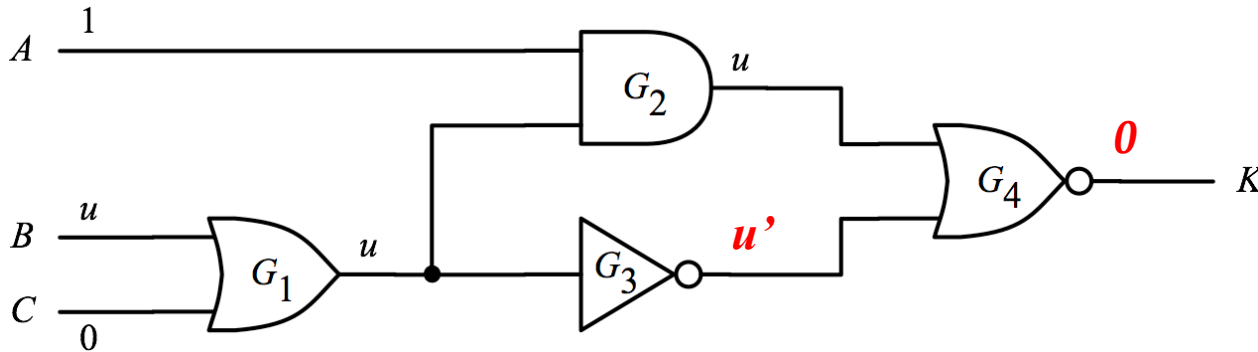
A, B, C are **Primary Inputs (PI)**
 $A_1 A_2$ are **Gate Inputs** for G_2
 K is **Primary Output (PO)**
 F is **Gate Output** for G_2



(b) Enumerate all possible cases ($B = 0$ and 1): $K = 0$

Possible Solution

- Introduce u' to represent inverse of u



- What is problem with this solution?
 - ♦ For every different source of u , need one **unique** u'
 - * Too many u 's !
 - * Need **symbolic simulation** to distinguish all u 's

**Moral: Simulation Results not 100% accurate.
Must Check!**

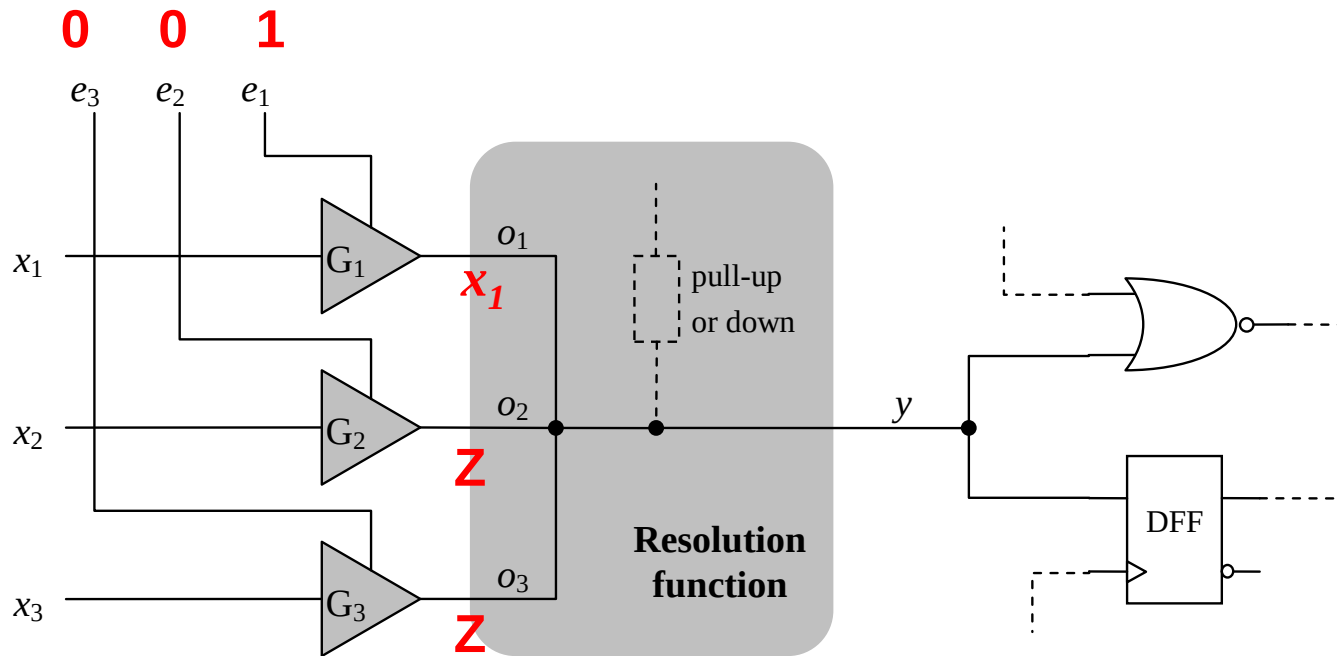
High Impedance State (Z)

- Tri-state bus driver

$$o_i = \begin{cases} x_i & \text{if } e_i = 1 \\ Z & \text{if } e_i = 0 \end{cases}$$

- Only one driver enabled at a time

♦ e_1, e_2, e_3 are **one-hot** signals

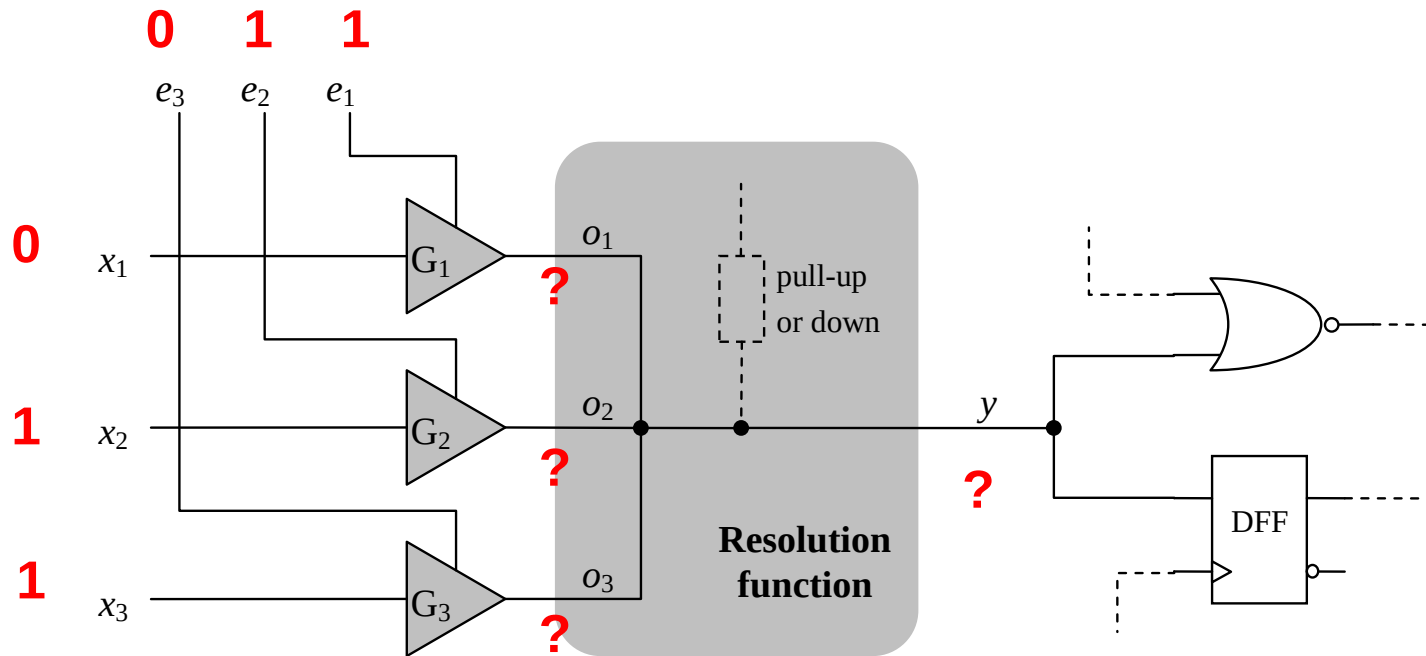


(WWW Fig 3.6)

Quiz

Q: If e_1, e_2, e_3 are NOT one-hot signals
what is y ?

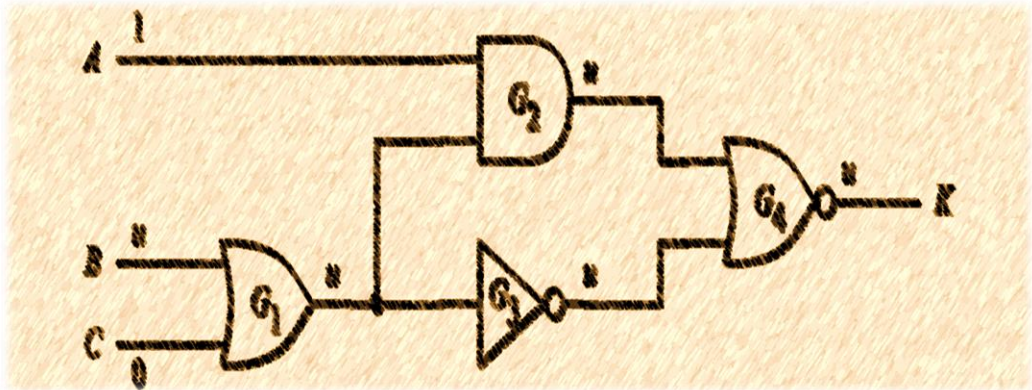
A:



(WWW Fig 3.6)

Logic Simulation

- Introduction
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 - * Truth table
 - * Input scanning
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Gate Evaluation Using Truth Table

- Determine output value based on a pre-stored truth table
- Example: 4-valued logic truth table

AND	0	1	u	z	OR	0	1	u	z	NOT	0	1	u	z
0	0	0	0	0	0	0	1	u	u		1	0	u	u
1	0	1	u	u	1	1	1	1	1					
u	0	u	u	u	u	u	1	u	u					
z	0	u	u	u	z	u	1	u	u					

Quiz:

What is size of truth table for n-input gate using 4 valued-logic?

A:

- Practical implementation
 - ◆ Break n-input gate into multiple 2-input gates
 - ◆ Use for-loop to evaluate gate one by one

Input Scanning

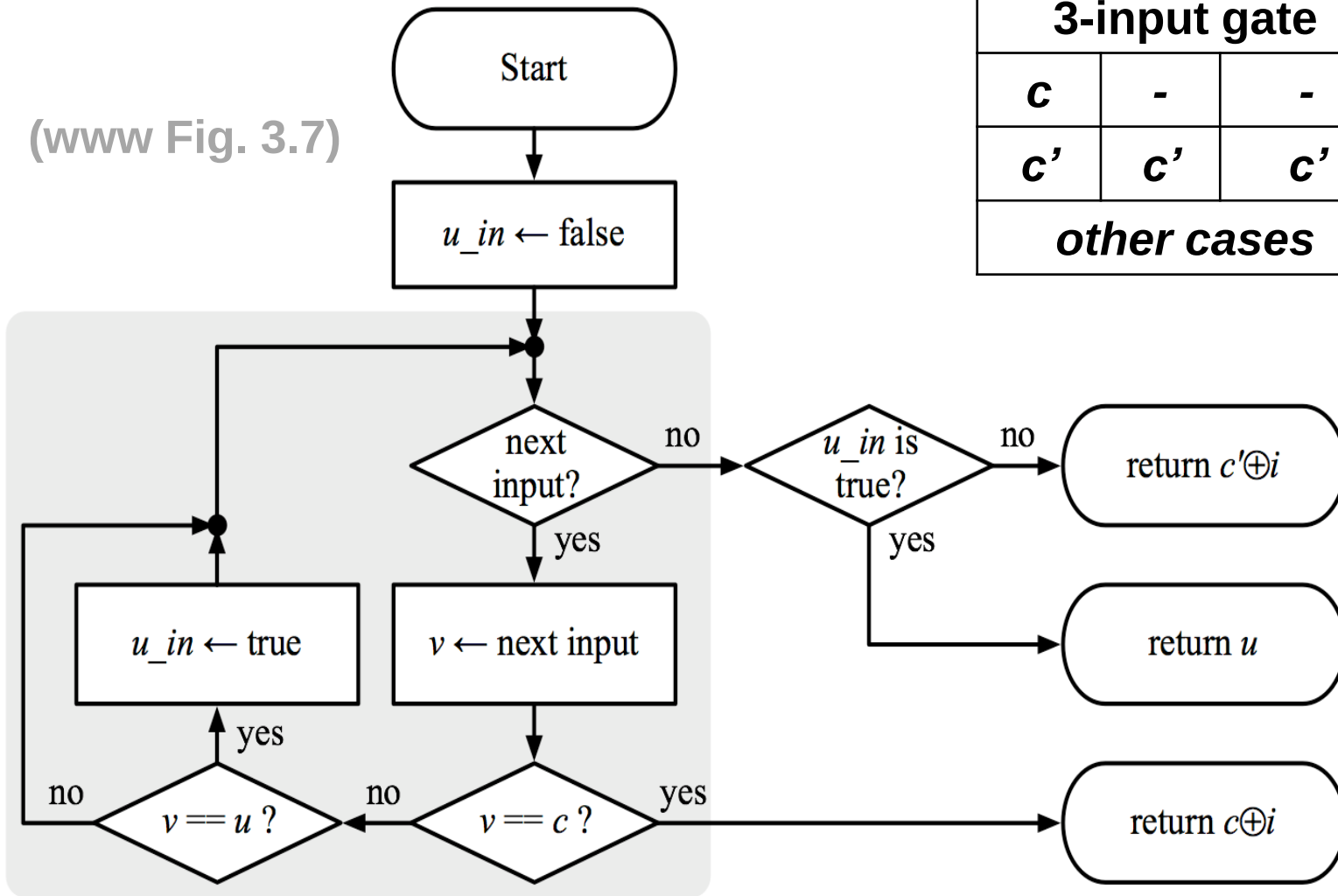
- Idea: check gate inputs one by one
 - ◆ Determine gate output by number of c, i, u
- Gates can be characterized by *
 - ◆ **Controlling value, c**
 - * c = input value that decides gate output
 - * regardless of other gate inputs
 - ◆ **Inversion, i**
 - * $i = 1$, gate output is inverted w.r.t. gate input
 - * $i = 0$, otherwise

	c	i
AND	0	0
OR	1	0
NAND	0	1
NOR	1	1

*Assumes only elementary gates :
AND, OR, NAND, and NOR

Input Scanning Algorithm

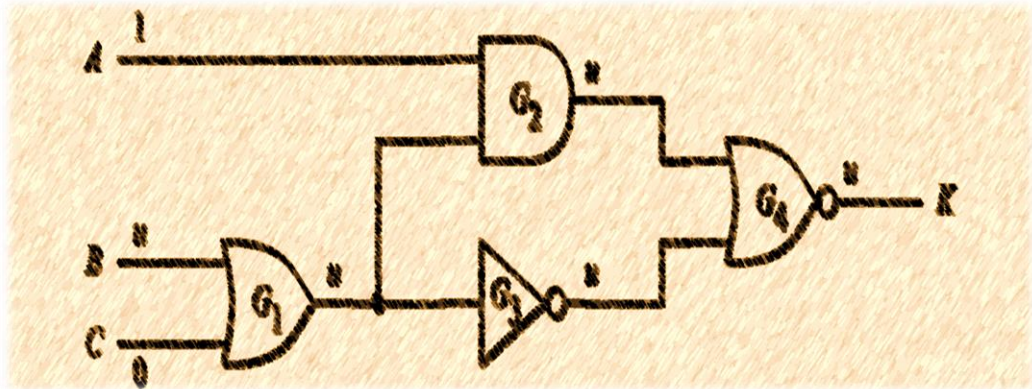
(www Fig. 3.7)



3-input gate			O/P
c	-	-	$c \oplus i$
c'	c'	c'	$c' \oplus i$
<i>other cases</i>			u

Logic Simulation

- Introduction
- Simulation Models
 - ◆ Logic States
 - ◆ Logic Gate Evaluation
 - ◆ Delay Models
 - * Gate delay model
 - * Wire delay (not in lecture)
- Logic Simulation Techniques
- Issues of Logic Simulations
- Conclusions

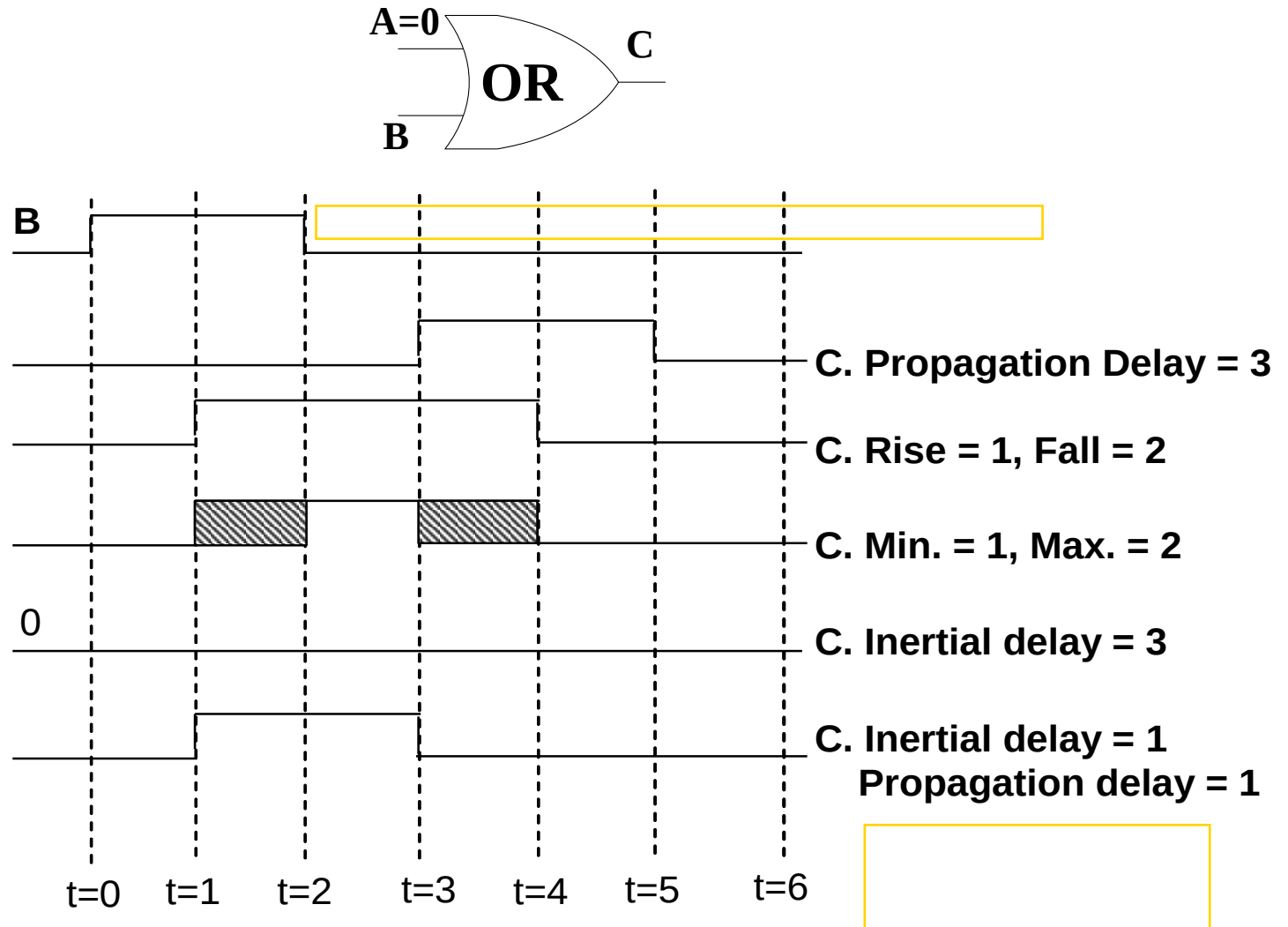


Gate Delay Models

- **Propagation delay (aka. Transport delay)**
 - ♦ DEF: time from gate input change to output change
 - * **Zero delay**: prop. delay of all gates is zero
 - * **Unit delay**: prop. delay of all gates is 1 time unit
 - * **Multiple delay**: prop. delay is multiple of some time unit
 - * **Rise delay, fall delay**
- **Ambiguous delay model**
 - ♦ **Max delay, Min delay, Typical (nominal) delay**
- **Inertial delay model**
 - ♦ Minimum amount of time during which a signal **must persist at gate input** in order to change gate output

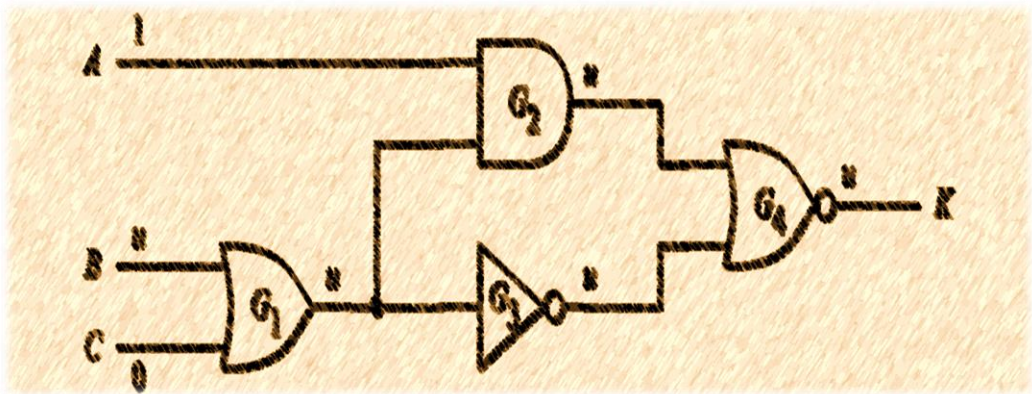
Different Delay Model, Different Usage

Example



Summary

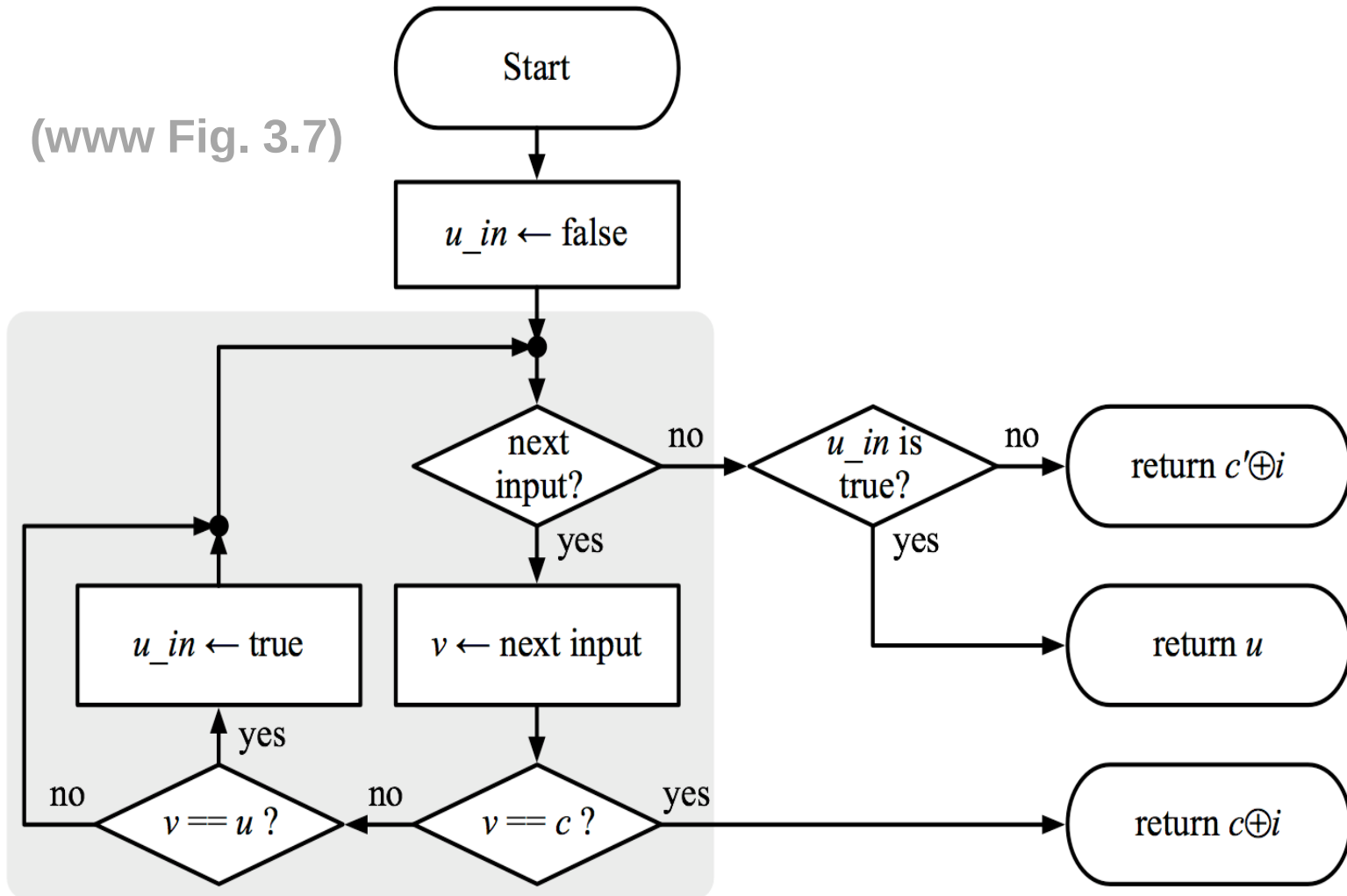
- Simulation Models
 - ◆ Logic States
 - * 0,1,u,z
 - ◆ Logic Gate Evaluation
 - * Truth table, Input scanning
 - ◆ Delay Models
 - * propagation, ambiguous, inertial delay



FFT 1

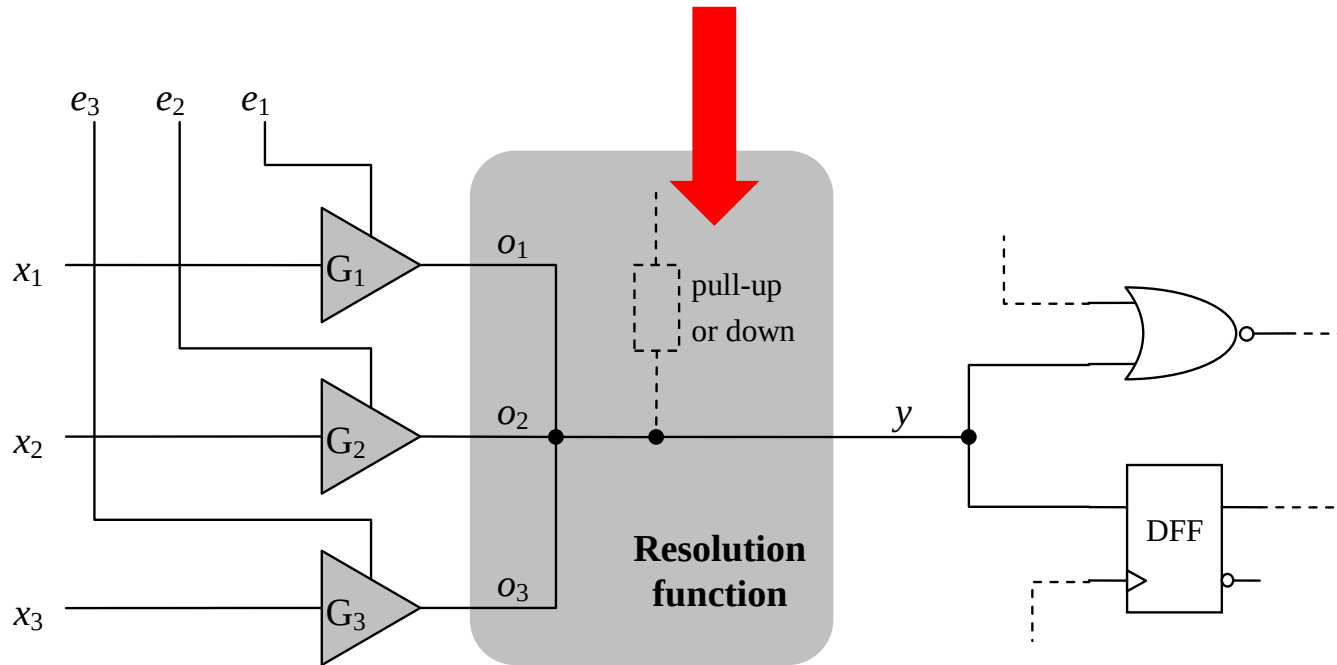
- Q: From truth table to input scanning, what do we gain/lose?
 - ♦ HINT: CPU time vs memory

(www Fig. 3.7)



FFT 2

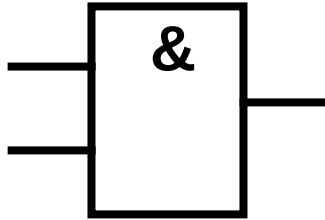
- Q: What is the use of pull-up or pull-down?
- HINT: what happens when $e_1=e_2=e_3=0$



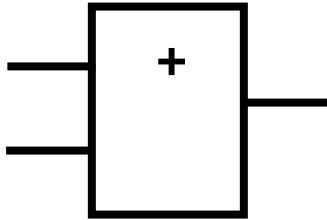
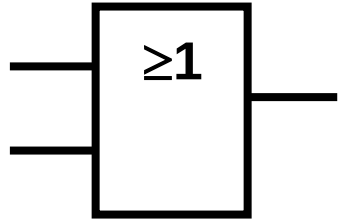
(WWW Fig 3.6)

Appendix: Logic Symbols

- IEEE logic symbols: rectangular shape v.s. distinctive shape
- AND



- Or



- inverter

